

**STRULE AUTOMATION**

Industrial Controls · Bay Area &amp; Northern California

**PLANT CHECKLIST**

# Crushing & Screening Controls Walk-Around Checklist

A one-page checklist to carry through the plant. It pairs with the full guide, [Getting More From a Crushing and Screening Plant Through Controls](#). Work top to bottom, note what you find, and fix the cheap honest things first. Nothing here asks you to defeat a safety function.

## Before You Start: Measure Utilization

- ☐ Pull run-time and stoppage logs from the PLC for a representative week
- ☐ Add up actual running hours at rate versus available hours
- ☐ List the top three stoppage causes by lost time, not by count
- ☐ Confirm the plant's rated tons per hour and compare to your best steady hour
- ☐ Decide: is this a capacity problem, or a utilization problem hiding as one?

## Feed Control and Choke Feeding

- ☐ Confirm the crusher runs choke fed (chamber full), not starved, most of the time
- ☐ Check what the feed loop controls on: motor amps or true power in kilowatts (kW)
- ☐ Find out whether the VFD or protection relay already publishes real power
- ☐ Watch the feeder for hunting or surging that points to an over-tight loop
- ☐ Slow a twitchy feed loop down before assuming you need more of anything

## Surge Bin and Buffer Management

- ☐ Note where the bin level normally sits: pinned full, pinned empty, or mid-range
- ☐ Confirm the regulating loop targets a mid-range band, not a hard number
- ☐ Look at the raw level signal for noise; add filtering if it bounces
- ☐ Check that the loop leans downstream rate gently rather than chasing level tightly
- ☐ Verify high- and low-level protection is separate logic from the regulating loop

## Screens: Blinding, Pegging, Recirculating Load

- ☐ Inspect decks for blinding (matted openings) and pegging (wedged particles)
- ☐ Note feed moisture and clay content, which make both worse
- ☐ Trend recirculating load if a scale is on the return; watch for a slow rise
- ☐ Trend screen motor power and any bearing or vibration signals over time
- ☐ Check product gradation drift as an early sign of a deck losing open area
- ☐ Ask whether a "crusher" or "feed" symptom is really a blinded deck upstream of the return

## Honest Measurement

- ☐ Zero the primary belt scale on an empty running belt
- ☐ Run a material test against a certified static (truck) weight
- ☐ Compare any two scales on the same stream; resolve disagreements before trusting either
- ☐ Sanity-check the power and level signals feeding your loops for drift or bad scaling
- ☐ Set a recurring calibration interval and actually keep it

## Trips and Protection (Never Defeat)

- ☐ List nuisance trips by machine: crusher power, belt drift, plugged chute, screen vibration
- ☐ For each, find the real mechanism before touching the setpoint
- ☐ Retune conservative trips to ride through normal surges while still protecting the machine
- ☐ Confirm no protective interlock is jumpered, bypassed, or widened past the point of protecting
- ☐ Leave every safety function intact; correct the cause, do not remove the guard

## When to Stop and Look at the Iron

- ☐ Screen media wrong for a sticky feed? That is media, stroke, or spray bars, not code
- ☐ Worn liner profile hurting product shape? That is a liner
- ☐ Chute plugs when wet? That is geometry
- ☐ A conveyor that cannot carry the rate? That is the belt, and it caps everything upstream
- ☐ Feed loop with no real-power signal available? A drive or power meter may be the honest fix

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Found more than a couple of these worth chasing and want an outside read? That is the work: walk the plant, separate the utilization losses from the real capacity limits, and say which is which. [struleautomation.com](http://struleautomation.com) or [info@struleautomation.com](mailto:info@struleautomation.com).